

Counting Independent Sets in a Co-comparability Graph

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Abstract

Our previous work (Lin and Chen, 2009) proposed some polynomial-time algorithms for counting independent sets (vertex covers) in a trapezoid graph. This paper extends that previous work. The previous results are extended to co-comparability graphs, which are a superclass of trapezoid graphs.

1 Introduction

A directed graph that is obtained from an undirected graph G by assigning a direction to each edge is called an *orientation* of G . A directed graph $G=(V,E)$ is *transitive* if $(u,v) \in E$ and $(v,w) \in E$ imply $(u,w) \in E$. An undirected graph is a *comparability graph* if it has a transitive orientation. A *co-comparability graph* is the complement of a comparability graph. Co-comparability graphs are usefully characterized by the co-comparability ordering. The *co-comparability ordering* is a numbering $1, 2, \dots, n$ of V such that if $u < v < w$ and $(u,w) \in E$, then either $(u,v) \in E$ or $(v,w) \in E$, or, equivalently, if $u < v < w$ and $(u,v), (v,w) \notin E$ then $(u,w) \notin E$. A graph G is a co-comparability graph if and only if G has a co-comparability ordering. Throughout this paper, vertices of a co-comparability graph are assumed to be numbered in a co-comparability ordering by $1, 2, \dots, n$.

An *independent set* (IS) is a subset of vertices of a graph such that there are no two vertices which are adjacent. Equivalently, each edge of the graph has at most one endpoint that lies in the IS. The *maximal independent set* (MIS) of a graph is an IS that is not a subset of any other IS in the graph. A set is called *minimum* (*maximum*) if it has the smallest (largest) size among all sets of its kind.

Let $G=(V,E)$ be a graph. A subset $C \subseteq V$ is called a *vertex cover* (VC) of G if and only if every edge in E has at least one endpoint in C . In graph theory, VC and IS are dual to each other. Since every edge in E is incident on a vertex in a VC C , no edge has neither endpoint in C . Thus, $V-C$ must be an IS.

This study investigates problems associated with the number of ISs. The number of ISs in a graph is important, particularly because it generally arises in problems that are related to network reliability [3,5]. In the art-gallery problem, the number of ISs is related to the probability that an art gallery is under complete surveillance when each guard has a given probability of falling asleep [5]. However, computing the number of ISs (VCs) is a $\#P$ -complete problem [4]. Valiant [6] defined the class of $\#P$ -complete problems. The class of $\#P$ problems includes problems that involve counting access computations for problems in NP , while the class of $\#P$ -complete problems includes the hardest problems in $\#P$. As is widely known, all exact algorithms for solving these problems have exponential time complexity, making efficient algorithms for this class of problems unlikely to be developed. However, this complexity can be reduced by considering only a restricted subclass of $\#P$ -complete problems.

Lin and Chen [3] presented $O(n^2)$ time algorithms for counting VCs (or ISs), minimal VCs (or MISs), minimum VCs (or maximum ISs), and maximum minimal VCs (or minimum MISs) in a trapezoid graph with n vertices. This paper extends these results to co-comparability graphs, which is a superclass of trapezoid graphs. Table 1 presents the results obtained in this paper.

Table 1. Summary of results;
 $O(n^\alpha)$ is the complexity of matrix multiplication,
 $\alpha < 2.3727$ [1,7].

Counting problems in a co-comparability graph	Time complexity
# of ISs	$O(n^2)$
# of MISs	$O(n^\alpha)$
# of maximum ISs	$O(n^2)$
# of minimum MISs	$O(n^\alpha)$

2 Preliminaries

This section presents the preliminaries on which the desired algorithms depend. Consider a co-comparability graph $G=(V,E)$ with vertices labeled in a co-comparability ordering, $V=\{1, 2, 3, \dots, n\}$. To simplify the implementation of the algorithm, two dummy vertices 0 and $n+1$ are

added to graph G .

Vertex u is said to be a predecessor of vertex v , denoted $u < v$, if $u < v$ and $(u, v) \notin E$. From the co-comparability ordering, if $u < v$ and $v < w$, then $u < w$. Let X be an $(n+2) \times (n+2)$ Boolean matrix of the relation $<$ on $V \cup \{0, n+1\}$, where $X(u, v) = 1$ if $u < v$ and $X(u, v) = 0$ otherwise. Given the adjacency matrices and the co-comparability ordering of a co-comparability graph with n vertices, X can be obtained in $O(n^2)$ time.

Vertex u is said to be an *immediate predecessor* of vertex v , denoted $u \ll v$, if $u < v$ and no vertex w exists such that $u < w < v$. Let Y be an $(n+2) \times (n+2)$ Boolean matrix of the relation $\ll$ on $V \cup \{0, n+1\}$, where $Y(u, v) = 1$ if $u \ll v$ and $Y(u, v) = 0$ otherwise. Consider the Boolean matrix product $X^2 = X \cdot X$ defined as

$$X^2(u, v) = \bigvee_{w=0}^{n+1} X(u, w) \wedge X(w, v).$$

Now $X^2(u, v) = 1$, which means that for some $w \in V$, $X(u, w) = 1$ and $X(w, v) = 1$; that is, $u < w$ and $w < v$. Therefore, by the definition of the immediate predecessor, Y can be expressed as $Y = X - X^2$. The bottleneck of the computation of Y is the pre-calculation of X^2 . The Boolean matrix product X^2 can be obtained by treating X as an integer matrix; computing the integer matrix product, and transforming the product matrix into a Boolean matrix. Let $O(n^\alpha)$ be the current best complexity of the algorithm for multiplying two $n \times n$ matrices. That $\alpha < 2.376$ [1] has long been known and this condition has been recently improved in unpublished work to $\alpha < 2.3727$ [7].

3 Counting independent sets

The section provides algorithms for counting ISs, MISs, maximum ISs and minimum MISs in a co-comparability graph. Given a graph G , the set of all ISs, MISs, maximum ISs and minimum MISs are denoted by $IS(G)$, $MIS(G)$, $maxIS(G)$ and $minMIS(G)$, respectively.

3.1. $O(n^2)$ -time algorithm for counting ISs

Let $N_G[v]$ denote the closed neighborhood of vertex v in G . The following lemma [3] is required for the proof in Theorem 1.

Lemma 1. [3] For a graph G , $|IS(G)| = |IS(G-v)| + |IS(G-N_G[v])|$ for each $v \in V$.

Given a co-comparability graph $G=(V, E)$, define $x(k) = \{i | i < k\} \equiv \{i | i < k \text{ and } (i, k) \notin E\}$, $0 \leq k \leq n+1$, as the set of all predecessors of vertex k . Whether $i \in x(k)$ can be easily determined by checking

whether $X(i, k) = 1$. For $1 \leq k \leq n+1$, let G_k be the subgraph of G that is induced by the vertex set $x(k) - \{0\}$. Accordingly, $G_k = G[x(k) - \{0\}]$. The following theorem can be used to compute the number of ISs.

Theorem 1. For $1 \leq k \leq n+1$,

$$|IS(G_k)| = \sum_{i \in x(k)} |IS(G_i)|, |IS(G_0)| = 1$$

Proof. Let $v = \max x(k)$. Clearly, $v < k$ and $(v, k) \notin E$. By Lemma 1,

$$|IS(G_k)| = |IS(G_k-v)| + |IS(G_k-N_G[v])|.$$

The present objective is to show $x(k) - N_G[v] = x(v)$. Since v is the largest vertex in $x(k)$, $x(k) - N_G[v] = \{i | i < v, (i, k) \notin E \text{ and } (i, v) \notin E\}$. By co-comparability ordering, when $i < v < k$ and $(v, k) \notin E$, $(i, v) \notin E$ implies $(i, k) \notin E$. Thus, $x(k) - N_G[v]$ can be rewritten as $\{i | i < v \text{ and } (i, v) \notin E\} = x(v)$. Hence, $|IS(G_k)| = |IS(G_k-v)| + |IS(G_v)|$.

Repeat the above process for the reduced co-comparability graph G_k-v until all vertices in $x(k)$ have been eliminated from G_k . Accordingly, Theorem 1 holds. $\square$

Based on Theorem 1, the following algorithm takes $O(n^2)$ time to compute the number of ISs in a co-comparability graph. Notably, the number of ISs in G can be obtained as $|IS(G_{n+1})|$ since $G_{n+1} = G$.

Algorithm 1. Compute the number of ISs in a co-comparability graph.

Compute X ; /* the Boolean matrix of the relation $<^*$ /

$|IS(G_0)| \leftarrow 1$;

for $k \leftarrow 1$ **to** $n+1$ **do**

$|IS(G_k)| \leftarrow 0$;

for $i \leftarrow 0$ **to** $k-1$ **do**

if $(X(i, k) = 1)$ **then** /* $i \in x(k)$ */

$|IS(G_k)| \leftarrow |IS(G_k)| + |IS(G_i)|$;

end for

end for

output("# of ISs is", $|IS(G_{n+1})|$);

end Algorithm

3.2. $O(n^2)$ -time algorithm for counting maximum ISs

Let $max_size(G)$ denote the largest size of ISs in G . Since $x(k)$ is the set of vertices of G in which every vertex is smaller than k and not adjacent to k , the following property holds.

Property 1. For $1 \leq k \leq n+1$,

$$max_size(G_k) = \max_{i \in x(k)} \{max_size(G_i) + 1\};$$

$$max_size(G_0) = 1.$$

By Property 1 and Theorem 1, the following algorithm takes $O(n^2)$ time to compute the number of maximum ISs in a co-comparability graph.

Algorithm 2. Compute the number of maximum ISs in a co-comparability graph.

```

Compute  $X$ ; /* the Boolean matrix of the relation
 $\prec$  */
 $|maxIS(G_0)| \leftarrow 1$ ;  $max\_size(G_0) \leftarrow 1$ ;
for  $k \leftarrow 1$  to  $n+1$  do
   $|maxIS(G_k)| \leftarrow 0$ ;  $max\_size(G_k) \leftarrow -\infty$ ;
  for  $i \leftarrow 0$  to  $k-1$  do
    if  $(X(i,k)=1)$  then begin /*  $i \in x(k)$  */
      if  $(max\_size(G_i)+1 > max\_size(G_k))$  then
         $|maxIS(G_k)| \leftarrow |maxIS(G_i)|$ ;
         $max\_size(G_k) \leftarrow max\_size(G_i)+1$ ;
      else if  $(max\_size(G_i)+1 = max\_size(G_k))$  then
         $|maxIS(G_k)| \leftarrow |maxIS(G_k)| + |maxIS(G_i)|$ ;
      end if
    end for
  end for
output (“# of maximum ISs is “,  $|maxIS(G_{n+1})|$ );
end Algorithm

```

3.3. $O(n^\alpha)$ -time algorithm for counting MISs ($\alpha < 2.3727$)

This section presents an algorithm for counting MISs in a co-comparability graph. Before the new algorithms are proposed, some properties of MISs and a theorem are presented here.

Let $MIS(G_k, i)$ be the set of all MISs in G_k that satisfy the condition for which i is the largest vertex in the MIS. Accordingly, $MIS(G_k, i) = \{I \mid I \in MIS(G_k) \text{ and } max\ I = i\}$. The following properties can be trivially established.

Property 2. For $i \neq j$, $MIS(G_k, i) \cap MIS(G_k, j) = \emptyset$.

Property 3. For $1 \leq k \leq n+1$,

$$MIS(G_k) = \bigcup_{i \in x(k)} MIS(G_k, i)$$

Define $y(k) = \{i \mid i \prec k\}$, $0 \leq k \leq n+1$, as the set of all immediate predecessors of vertex k . Whether $i \in y(k)$ can be easily determined by checking whether $Y(i, k) = 1$.

Property 4. For $1 \leq k \leq n+1$, $y(k)$ is a clique.

Proof. For each pair $i, j \in y(k)$, by definition of $y(k)$, $i \prec x(j)$ and $j \prec x(i)$. Hence, $(i, j) \in E$. $\square$

Property 5. For $i \in x(k)$ and $i \notin y(k)$, $MIS(G_k, i) = \emptyset$.

Proof. $i \in x(k)$ and $i \notin y(k)$ imply that there exists $j \in x(k)$ such that $i \prec j \prec k$. Suppose that there exists an MIS $I \in MIS(G_k, i)$. Clearly, $I \cup \{j\}$ is also an IS

in G_k , which contradicts the fact that I is an MIS in G_k . $\square$

Property 6. For $i \in y(k)$,

$$MIS(G_k, i) = \{I' \cup \{i\} \mid I' \in MIS(G_i)\}.$$

Proof. ($\Leftarrow$) Let I' be an MIS in G_i . Since $i \in y(k)$, there exists no j such that $i \prec j \prec k$. Therefore, $I' \cup \{i\}$ is an MIS in G_k .

($\Rightarrow$) Let $I \in MIS(G_k, i)$ and $I' = I - \{i\}$. Since $i \in y(k)$, all $v \in I'$, $v \prec i$ and I' is an IS in G_i . Suppose there exists $j \neq i$ such that $I' \cup \{j\}$ is an IS in G_i . Then, $I' \cup \{j\} \cup \{i\}$ is an IS in G_k , i.e. $I \cup \{j\}$ is also an IS in G_k . This contradicts the fact that I is an MIS in G_k . $\square$

By properties 2, 3, 5, and 6, the following theorem holds.

Theorem 2. For $1 \leq k \leq n+1$,

$$|MIS(G_k)| = \sum_{i \in y(k)} |MIS(G_i)|, |MIS(G_0)| = 1.$$

From Theorem 2, a straightforward algorithm can be derived to compute the number of MISs in a co-comparability graph as follows.

Algorithm 3. Compute the number of MISs in a co-comparability graph.

```

Compute  $Y$ ; /* the Boolean matrix of the relation
 $\ll$  */
 $|MIS(G_0)| \leftarrow 1$ ;
for  $k \leftarrow 1$  to  $n+1$  do
   $|MIS(G_k)| \leftarrow 0$ ;
  for  $i \leftarrow 0$  to  $k-1$  do
    if  $(Y(i,k)=1)$  then /*  $i \in y(k)$  */
       $|MIS(G_k)| \leftarrow |MIS(G_k)| + |MIS(G_i)|$ ;
    end for
  end for
output (“# of maximal ISs is “,  $|MIS(G_{n+1})|$ );
end Algorithm

```

As mentioned in the Preliminaries section, the Boolean matrix of the relation $\ll$, Y , can be computed in $O(n^{2.3727})$ time. Thus, Algorithm 3 also takes $O(n^{2.3727})$ time.

3.4. $O(n^\alpha)$ -time algorithm for counting minimum MISs ($\alpha < 2.3727$)

Let $minM_size(G)$ be the smallest size of MISs in G . The following property is similar to Property 1 but takes into account the smallest value for all $i \in y(k)$.

Property 7. For $1 \leq k \leq n+1$,

$$minM_size(G_k) = \min_{i \in y(k)} \{minM_size(G_i) + 1\};$$

$$minM_size(G_0) = 1.$$

By Property 7 and Theorem 2, the following algorithm takes $O(n^{2.3727})$ time to compute the number of minimum MISs in a co-comparability graph.

Algorithm 4. Compute the number of minimum MISs in a co-comparability graph.

```

Compute  $Y$ ; /* the Boolean matrix of the relation
 $\ll$  */
 $|minMIS(G_0)| \leftarrow 1$ ;  $minM\_size(G_0) \leftarrow 1$ ;
for  $k \leftarrow 1$  to  $n+1$  do
   $|minMIS(G_k)| \leftarrow 0$ ;  $minM\_size(G_k) \leftarrow +\infty$ ;
  for  $i \leftarrow 0$  to  $k-1$  do
    if ( $Y(i,k)=1$ ) then begin /*  $i \in y(k)$  */
      if ( $minM\_size(G_i)+1 < minM\_size(G_k)$ ) then
         $|minMIS(G_k)| \leftarrow |minMIS(G_i)|$ ;
         $minM\_size(G_k) \leftarrow minM\_size(G_i)+1$ ;
      else if ( $minM\_size(G_i)+1 = minM\_size(G_k)$ ) then
         $|minMIS(G_k)| \leftarrow |minMIS(G_i)| + |minMIS(G_i)|$ ;
    end if
  end for
end for
output("The number of minimum maximal
independent sets is ",  $|minMIS(G_{n+1})|$ );
end Algorithm

```

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